

Consider the **steady buoyancy-free** motion of a perfect liquid of density ρ , viscosity μ , thermal conductivity k , and specific heat c , induced by an infinitely long porous cylinder of radius R that moves with a constant velocity U_w parallel to its axis, while a constant flow rate $q = 2\pi R V_w$ per unit length is uniformly sucked in the radial direction through the porous wall of the cylinder. Far from the cylinder, $r \rightarrow \infty$, the fluid is at rest, at pressure p_∞ and temperature T_∞ .

To solve the problem, use a cylindrical coordinate system (r, θ, z) , where z is the axis of the cylinder, and assume that the velocity field is axisymmetric, $\frac{\partial \bar{v}}{\partial \theta} = 0$, as well as fully developed, $\frac{\partial \bar{v}}{\partial z} = 0$, thus $\bar{v} = v_r(r)\bar{e}_r + v_z(r)\bar{e}_z$.

1. Using continuity, determine the velocity component $v_r(r)$. (1 point)
2. Obtain the pressure field $p(r, z)$ by integration of the r -momentum equation, demonstrating in particular that $\frac{\partial p}{\partial z} = 0$. (2 points)
3. Using the z -momentum equation, obtain the axial velocity distribution, $v_z(r)$. (2 points)
4. Compute the trajectory of a particle initially placed at $r = R_i > R$ and $z = z_i$, as well as the corresponding sucking time, t_s . (1 point)
5. Calculate the force per unit length that the liquid exerts on the cylinder. (1 point)
6. Write the differential equation that determines the temperature in the liquid $T(r)$. Obtain the function $T(r)$, assuming that the wall is at constant uniform temperature T_w . (2 points)
7. Obtain the conductive heat flux per unit length from the liquid to the wall q_w . (1 point)
8. Show that there is no steady solution if fluid is blown through the wall, instead of being sucked. (1 extra point)